



## Time Series Analysis

### Chapter 10: State space models, Kalman filter and Markov switching



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## Acronyms used in this material (1/2)

- ▣ ACF: Autocorrelation Function
- ▣ AI: Artificial Intelligence
- ▣ AIC: Akaike Information Criterion
- ▣ AR: AutoRegressive (model); in event studies: Abnormal Return
- ▣ ARCH: AutoRegressive Conditional Heteroskedasticity
- ▣ ARIMA: AutoRegressive Integrated Moving Average (model)
- ▣ ARMA: AutoRegressive Moving Average (model)
- ▣ BET: Bucharest Exchange Trading (index)
- ▣ BIC: Bayesian Information Criterion
- ▣ EM: Expectation–Maximisation (algorithm)
- ▣ EODHD: EOD Historical Data (market-data provider)
- ▣ ETS: Error, Trend, Seasonality (exponential smoothing state space models)
- ▣ EU: European Union
- ▣ FRED: Federal Reserve Economic Data
- ▣ GARCH: Generalised AutoRegressive Conditional Heteroskedasticity
- ▣ GDP: Gross Domestic Product
- ▣ GDPC1: Real Gross Domestic Product, chained dollars (FRED series code)
- ▣ HP: Hodrick–Prescott (filter)
- ▣ MA: Moving Average
- ▣ ML: Machine Learning
- ▣ MS-AR: Markov-Switching AutoRegressive (model)
- ▣ NASA: National Aeronautics and Space Administration
- ▣ NBER: National Bureau of Economic Research
- ▣ OLS: Ordinary Least Squares
- ▣ RTS: Rauch–Tung–Striebel (smoother)
- ▣ SD: Standard Deviation

## Acronyms used in this material (2/2)

- **SES: Simple Exponential Smoothing**
- **SWARCH: Switching ARCH**
- **TVP: Time-Varying Parameter (regression)**
- **UC: Unobserved Components (model)**
- **US: United States**
- **USREC: US Recession indicator (FRED series code)**
- **VAR: Vector AutoRegression**



## Today's question and route

- **Question:** how do we measure what we cannot observe directly: the true level of a noisy series, the output gap, a changing beta, a recession?
  - ▶ the answer: write the unobserved quantity as a **state** and let the data update our estimate of it, period by period
- **Route** of the chapter
  - ▶ the state space form: measurement and transition equations; local level, local linear trend, ARMA, time-varying regression
  - ▶ the Kalman filter by hand; smoothing; likelihood and estimation; missing values
  - ▶ trend and cycle: Romanian GDP, the HP filter and its critique; dynamic factors and nowcasting
  - ▶ Markov switching: recessions, Romanian growth regimes, volatility regimes
- We build on [Chapter 0](#) (exponential smoothing), [Chapter 2](#) (ARMA), [Chapter 5](#) (GARCH) and [Chapter 8](#) (long memory); Seminar 10 comes before this lecture

## Learning outcomes

- Write a model in state space form: local level, local linear trend, AR(2), a regression with a time-varying coefficient
- Run the Kalman filter by hand and explain the Kalman gain
- Show that simple exponential smoothing is the steady state of the local level filter
- Estimate a state space model by maximum likelihood, smooth it, and handle missing values and forecasts
- Decompose GDP into trend and cycle and judge the HP filter against the critique of Hamilton (2018)
- Estimate a Markov-switching model and read its regime probabilities, durations and pitfalls

## Reading and tools

- Textbook: [Huang and Petukhina \(2022\)](#), the chapter on state space models and the Kalman filter
  - ▶ state space: [Durbin and Koopman \(2012\)](#) (the Nile example of this chapter comes from their book); [Harvey \(1989\)](#); [Shumway and Stoffer \(2017\)](#), Ch. 6
  - ▶ regime switching: [Hamilton \(1989\)](#); [Hamilton \(1994\)](#), Ch. 13 (Kalman filter) and Ch. 22 (regime changes); [Kim and Nelson \(1999\)](#)
- Python Quantlets of this chapter: [Quantlets/Ch\\_10](#)
  - ▶ a Kalman filter written out in `numpy`; `statsmodels`: `UnobservedComponents`, `SARIMAX`, `DynamicFactor`, `MarkovRegression`, `MarkovAutoregression`
- Lecture notebook: [open in Google Colab](#)
- Video course:  [Kalman Filter](#)

## Rudolf Kálmán and the Apollo navigation



Rudolf E. Kálmán (1930–2016)

Photo: ETH-Bibliothek Zürich, Bildarchiv (2007); CC BY-SA 4.0; Wikimedia Commons



Apollo 8 (1968): Jim Lovell at the guidance and navigation station

Photo: NASA (1968); public domain; Wikimedia Commons

- **Kalman (1960)**: a recursive estimate of a hidden state from noisy measurements
  - ▶ at each step: predict the state, then correct the prediction with the new measurement
  - ▶ the correction weight, the **Kalman gain**, balances the two sources of uncertainty
- The filter was adapted at NASA Ames for the navigation of the Apollo spacecraft
  - ▶ state: position and velocity; measurements: star sightings and radar
- In economics the state is a trend, a cycle, a coefficient or a regime; the measurements are the published data

## Hidden states in economics and finance

| Observed series              | Hidden state                              | Model in this chapter   |
|------------------------------|-------------------------------------------|-------------------------|
| Nile flow, monthly inflation | the underlying level                      | local level             |
| real GDP                     | trend and output gap                      | unobserved components   |
| stock index returns          | a time-varying beta                       | time-varying regression |
| several activity indicators  | one common factor                         | dynamic factor model    |
| GDP growth, returns          | recession or expansion; calm or turbulent | Markov switching        |

- The first four have a **continuous** state and are filtered by the Kalman filter
- The last has a **discrete** state (a regime) and is filtered by the Hamilton filter
- Both filters do the same thing: predict the state, then update the prediction with  $y_t$

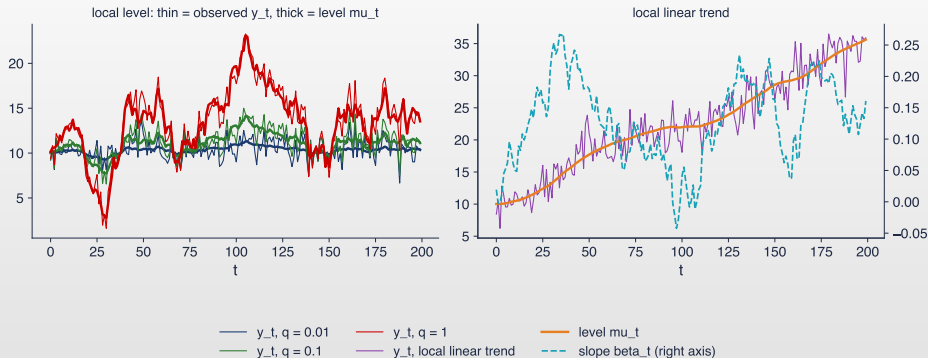
## The state space form

- ▣ **Measurement equation:**  $y_t = Z_t \alpha_t + \varepsilon_t$ ,  $\varepsilon_t \sim N(0, H)$ 
  - ▶  $y_t$ : the observation (here a scalar);  $\alpha_t$ : the **state vector** ( $m \times 1$ ), not observed;  $Z_t$ : a  $1 \times m$  row of known numbers
- ▣ **Transition equation:**  $\alpha_{t+1} = T \alpha_t + R \eta_t$ ,  $\eta_t \sim N(0, Q)$ 
  - ▶ the state evolves as a first-order vector autoregression (VAR(1), [Chapter 6](#));  $T$  is  $m \times m$
  - ▶  $R$ : maps the shocks  $\eta_t$  into the state;  $H$ ,  $Q$ : the variances of the measurement and of the state shocks
  - ▶ initial state:  $\alpha_1 \sim N(a_1, P_1)$ ,  $a_1$ ,  $P_1$ : its mean and variance;  $\varepsilon_t$ ,  $\eta_s$  and  $\alpha_1$  mutually independent
- ▣ The **system matrices**  $Z_t$ ,  $H$ ,  $T$ ,  $R$ ,  $Q$  contain the parameters  $\theta$ ; they are estimated by maximum likelihood
  - ▶ the same form covers ARMA, exponential smoothing, trend-cycle models, regressions with moving coefficients and factor models [Durbin and Koopman \(2012\)](#)

## The local level model

- **Local level** (random walk plus noise):  $y_t = \mu_t + \varepsilon_t$ ,  $\mu_{t+1} = \mu_t + \eta_t$ 
  - ▶ state  $\alpha_t = \mu_t$ , the level;  $Z = T = R = 1$ ,  $H = \sigma_\varepsilon^2$ ,  $Q = \sigma_\eta^2$
  - ▶  $\varepsilon_t$ : measurement noise that disappears next period;  $\eta_t$ : a permanent shift of the level
- **Signal-to-noise ratio**  $q = \sigma_\eta^2 / \sigma_\varepsilon^2$ : how much of the movement is permanent
  - ▶  $q = 0$ : a constant level (white noise around a mean);  $q \rightarrow \infty$ : a pure random walk
- **Reduced form**:  $\Delta y_t = \eta_{t-1} + \varepsilon_t - \varepsilon_{t-1}$  is an MA(1) with  $\rho(1) = -1/(q+2)$ 
  - ▶ so the local level model is an ARIMA(0,1,1) ([Chapter 3](#)) with a negative MA coefficient
  - ▶  $\gamma(0) = \sigma_\eta^2 + 2\sigma_\varepsilon^2$  and  $\gamma(1) = -\sigma_\varepsilon^2$  give  $\rho(1)$

## Local level and local linear trend paths



- Left: the same shocks  $\varepsilon_t, \eta_t \sim N(0, 1)$  with  $\sigma_\varepsilon^2 = 1$  and  $q = 0.01; 0.1; 1$  (thin:  $y_t$ , thick:  $\mu_t$ ). Right: a local linear trend, whose slope  $\beta_t$  is itself a random walk

## Interpreting the simulated paths

- Small  $q$ : the level hardly moves and  $y_t$  looks like noise around a mean; large  $q$ : the level wanders and  $y_t$  follows it
  - ▶ first-difference autocorrelation:  $-0.51$  for  $q = 0.01$  (theory  $-1/2.01$ ) and  $-0.28$  for  $q = 1$  (theory  $-1/3$ )
- The local linear trend has a slope that changes slowly: periods of fast and slow growth
  - ▶ this is the trend model we will use for GDP
- The estimation problem: from  $y_t$  alone, how much is level and how much is noise? The Kalman filter answers it for given variances; the likelihood chooses the variances

## The local linear trend

- **Local linear trend:**  $y_t = \mu_t + \varepsilon_t$ ,  $\mu_{t+1} = \mu_t + \beta_t + \eta_t$ ,  $\beta_{t+1} = \beta_t + \zeta_t$ 
  - ▶ state  $\alpha_t = (\mu_t, \beta_t)'$ : the level and the slope (the growth rate)
  - ▶  $Z = (1, 0)$ ,  $T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ ,  $Q = \text{diag}(\sigma_\eta^2, \sigma_\zeta^2)$
- Special cases
  - ▶  $\sigma_\eta^2 = \sigma_\zeta^2 = 0$ : a deterministic linear trend  $\mu_1 + \beta_1 t$  ([Chapter 3](#))
  - ▶  $\sigma_\zeta^2 = 0$ : a random walk with drift  $\beta$
  - ▶  $\sigma_\eta^2 = 0$ : the **smooth trend** (integrated random walk), the model behind the HP filter ([Section 5](#))
- Forecasts:  $\hat{y}_{T+h} = \hat{\mu}_T + h\hat{\beta}_T$ , the state space version of Holt's method ([Chapter 0](#))

## ARMA models in state space form

- **AR(2)**:  $y_t = \phi_1 y_{t-1} + \phi_2 y_{t-2} + u_t$ ; state  $\alpha_t = (y_t, y_{t-1})'$ 
  - ▶  $y_t = (1, 0) \alpha_t$  ( $H = 0$ ),  $\alpha_{t+1} = \begin{pmatrix} \phi_1 & \phi_2 \\ 1 & 0 \end{pmatrix} \alpha_t + \begin{pmatrix} 1 \\ 0 \end{pmatrix} u_{t+1}$
  - ▶  $T$  is the companion matrix of [Chapter 2](#); its eigenvalues are the inverse AR roots
- **ARMA(1,1)**:  $y_t = \phi y_{t-1} + u_t + \theta u_{t-1}$  with  $\alpha_t = (y_t, \theta u_t)'$ 
  - ▶  $T = \begin{pmatrix} \phi & 1 \\ 0 & 0 \end{pmatrix}$ ,  $R = (1, \theta)'$ ; any ARMA( $p, q$ ) fits with  $m = \max(p, q + 1)$
- Why it matters: the Kalman filter gives the **exact** Gaussian likelihood of an ARMA model, with missing values allowed
  - ▶ the stationary start:  $P_1$  solves  $P = TPT' + RQR'$ , i.e.  $\text{vec}(P_1) = (I - T \otimes T)^{-1} \text{vec}(RQR')$
  - ▶ this is how `statsmodels` SARIMAX estimates the ARIMA models of [Chapters 2–4](#)

## Worked example: AR(2) for US GDP growth

- ▣ Data: US real GDP growth, quarter on quarter, 1947–2019 ( $T = 291$ , FRED GDPC1)
  - ▶ AR(2) with a constant by SARIMAX:  $\hat{\phi}_1 = 0.319$ ,  $\hat{\phi}_2 = 0.111$ ,  $\hat{\sigma}^2 = 0.739$ , mean  $\hat{\mu} = 0.774\%$  per quarter
- ▣ Our own filter, with the state space matrices of the previous slide and the stationary  $P_1$ 
  - ▶ log-likelihood: Kalman  $-368.9157$ ; SARIMAX  $-368.9157$ : the same number
- ▣ The prediction errors  $v_t$  of the filter are the one-step forecast errors of the AR(2); the likelihood is built from them (Section 4)

## Regression with time-varying parameters

- **Time-varying parameter (TVP) regression:**  $y_t = \alpha + \beta_t x_t + \varepsilon_t$ ,  $\beta_{t+1} = \beta_t + \eta_t$ 
  - ▶ state  $(\alpha, \beta_t)'$ ; the row  $Z_t = (1, x_t)$  **changes over time**;  $Q = \text{diag}(0, \sigma_\eta^2)$
  - ▶  $\sigma_\eta^2 = 0$ : ordinary least squares (OLS), computed recursively
- Examples
  - ▶ a market beta that rises in crises; the pass-through of the exchange rate to inflation; the persistence of inflation
- Better than a rolling window: no window length to choose, every observation is used, and  $\beta_t$  comes with a standard error

## Exponential smoothing as a state space model

- **Chapter 0:** simple exponential smoothing (SES) updates a level  $\ell_t = \alpha y_t + (1 - \alpha)\ell_{t-1}$  and forecasts  $\hat{y}_{t+1} = \ell_t$ 
  - ▶ equivalently  $\ell_t = \ell_{t-1} + \alpha(y_t - \ell_{t-1})$ : old level plus a share  $\alpha$  of the surprise
- **Muth (1960):** SES gives the optimal forecasts when the data follow the local level model
  - ▶ Section 3 shows it:  $\alpha$  is the steady-state Kalman gain
- ETS models (error, trend, seasonal) are state space models with a single source of error [Hyndman et al. \(2008\)](#); the models here have separate shocks for each component

## Recap: The state space form

- ▣  $y_t = Z_t \alpha_t + \varepsilon_t$ ,  $\alpha_{t+1} = T \alpha_t + R \eta_t$ : a hidden VAR(1) observed with noise
- ▣ Local level:  $q = \sigma_\eta^2 / \sigma_\varepsilon^2$ ; reduced form ARIMA(0,1,1)
- ▣ Local linear trend, ARMA and TVP regression are other choices of  $Z_t$ ,  $T$ ,  $Q$
- ▣ The Kalman likelihood of an AR(2) equals the exact ARMA likelihood

## The idea of the filter

- Notation:  $Y_t = (y_1, \dots, y_t)$ ; we track two numbers (vectors) for the state
  - ▶ prediction:  $a_t = E(\alpha_t \mid Y_{t-1})$  with variance  $P_t$
  - ▶ filtered estimate:  $a_{t|t} = E(\alpha_t \mid Y_t)$  with variance  $P_{t|t}$
- Each period has two steps
  - ▶ **update**:  $y_t$  arrives; correct  $a_t$  by a share of the surprise  $v_t = y_t - Z_t a_t$
  - ▶ **prediction**: push the filtered state through the transition equation to get  $a_{t+1}$
- With Gaussian errors,  $a_{t|t}$  is the exact conditional mean; without normality it is still the best **linear** estimate  
[Durbin and Koopman \(2012\)](#)
- Start:  $a_1, P_1$ ; for a non-stationary state we use a **diffuse** start ( $P_1$  very large: we know nothing)

## The two steps in formulas

### □ **Update** (when $y_t$ is observed)

- ▶ prediction error  $v_t = y_t - Z_t a_t$ , with variance  $F_t = Z_t P_t Z_t' + H$
- ▶ **Kalman gain**  $K_t = P_t Z_t' F_t^{-1}$
- ▶  $a_{t|t} = a_t + K_t v_t$ ,  $P_{t|t} = P_t - K_t Z_t P_t$

### □ **Prediction**

- ▶  $a_{t+1} = T a_{t|t}$ ,  $P_{t+1} = T P_{t|t} T' + R Q R'$

### □ **Local level:** $F_t = P_t + \sigma_\varepsilon^2$ , $K_t = P_t / (P_t + \sigma_\varepsilon^2)$

- ▶  $a_{t+1} = a_t + K_t (y_t - a_t)$ ,  $P_{t+1} = P_t (1 - K_t) + \sigma_\eta^2$

### □ Missing $y_t$ : skip the update ( $a_{t|t} = a_t$ , $P_{t|t} = P_t$ ) and predict

## The Kalman gain

- Local level:  $K_t = \frac{P_t}{P_t + \sigma_\varepsilon^2}$  is the weight of the new observation, between 0 and 1
  - ▶  $a_{t|t} = (1 - K_t) a_t + K_t y_t$ : a weighted average of the prediction and the observation
- Two uncertainties compete
  - ▶ noisy measurements (large  $\sigma_\varepsilon^2$ ): small  $K_t$ , trust the model
  - ▶ uncertain state (large  $P_t$ , e.g. at the start or after missing data): large  $K_t$ , trust the data
- The variance always falls with an observation:  $P_{t|t} = (1 - K_t)P_t \leq P_t$ 
  - ▶ and grows by  $\sigma_\eta^2$  in the prediction step: the two forces settle at a steady state
- The same formula as the weighting of two independent estimates by their precisions  $1/P_t$  and  $1/\sigma_\varepsilon^2$

## Worked example: three periods by hand (1/2)

- Local level with  $\sigma_\varepsilon^2 = 4$ ,  $\sigma_\eta^2 = 1$  ( $q = 0.25$ ); prior  $a_1 = 10$ ,  $P_1 = 4$ ; data  $y = (12, 11, 14)$ 
  - ▶  $t = 1$ :  $F_1 = 4 + 4 = 8$ ,  $K_1 = 4/8 = 0.5$ ,  $v_1 = 12 - 10 = 2$
  - ▶  $a_{1|1} = 10 + 0.5 \cdot 2 = 11$ ,  $P_{1|1} = 4(1 - 0.5) = 2$ ; prediction  $a_2 = 11$ ,  $P_2 = 2 + 1 = 3$
- $t = 2$ :  $F_2 = 3 + 4 = 7$ ,  $K_2 = 3/7 = 0.429$ ,  $v_2 = 11 - 11 = 0$ 
  - ▶ no surprise: the level stays at  $a_{2|2} = 11$ , but the uncertainty falls to  $P_{2|2} = 3 \cdot 4/7 = 1.71$
  - ▶ prediction  $a_3 = 11$ ,  $P_3 = 1.71 + 1 = 2.71$
- The gain falls from 0.5 to 0.429: the filter learns about the level and relies less on each new observation

## Worked example: three periods by hand (2/2)

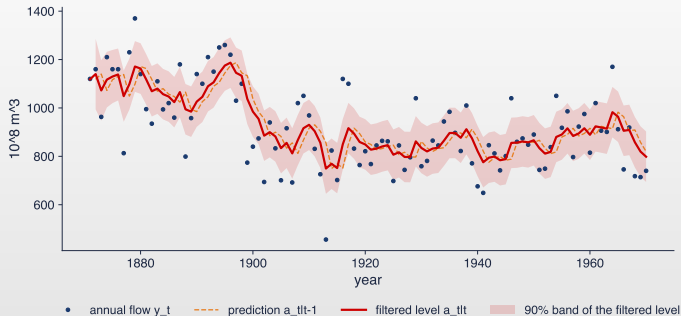
| $t$ | $y_t$ | $a_t$ | $P_t$ | $F_t$ | $K_t$ | $v_t$ | $a_{t t}$ | $P_{t t}$ |
|-----|-------|-------|-------|-------|-------|-------|-----------|-----------|
| 1   | 12    | 10.00 | 4.00  | 8.00  | 0.500 | 2     | 11.00     | 2.00      |
| 2   | 11    | 11.00 | 3.00  | 7.00  | 0.429 | 0     | 11.00     | 1.71      |
| 3   | 14    | 11.00 | 2.71  | 6.71  | 0.404 | 3     | 12.21     | 1.62      |

- $t = 3$ :  $K_3 = 2.71/6.71 = 0.404$ ,  $v_3 = 14 - 11 = 3$ ,  $a_{3|3} = 11 + 0.404 \cdot 3 = 12.21$ 
  - forecast for  $t = 4$ :  $a_4 = 12.21$  with  $P_4 = 2.62$ ; the forecast of  $y_4$  has variance  $P_4 + \sigma_\varepsilon^2$
- The gain keeps falling towards the steady state  $\bar{K} = 0.390$  ( $\bar{P} = 2.56$ , next slide)
  - the surprise of 3 moves the level by only  $0.404 \cdot 3$ : part of it is treated as noise

## Steady state and exponential smoothing

- For the local level model,  $P_t$  converges to the solution of  $\bar{P} = \bar{P}(1 - \bar{K}) + \sigma_\eta^2$  with  $\bar{K} = \bar{P}/(\bar{P} + \sigma_\varepsilon^2)$ 
  - ▶ solution:  $\bar{P} = \sigma_\varepsilon^2 \frac{q + \sqrt{q^2 + 4q}}{2}$ ,  $\bar{K} = \frac{\bar{P}}{\bar{P} + \sigma_\varepsilon^2}$
  - ▶ example:  $q = 0.25$  gives  $\bar{P}/\sigma_\varepsilon^2 = (0.25 + \sqrt{1.0625})/2 = 2.56/4$  and  $\bar{K} = 0.390$
- In the steady state:  $a_{t+1} = a_t + \bar{K}(y_t - a_t) = \bar{K}y_t + (1 - \bar{K})a_t$ 
  - ▶ this is SES with  $\alpha = \bar{K}$ : exponential smoothing is the Kalman filter of a local level model [Muth \(1960\)](#)
- Inverse map: an SES weight  $\alpha$  corresponds to  $q = \alpha^2/(1 - \alpha)$ 
  - ▶ the state space version adds what SES lacks: a likelihood for  $q$ , forecast intervals, missing values

## The Nile: the Kalman filter of the local level model

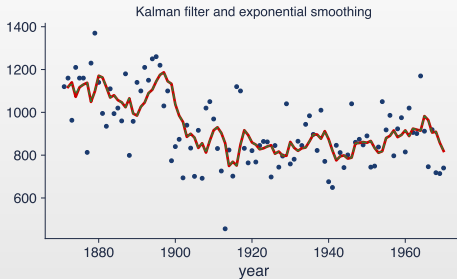
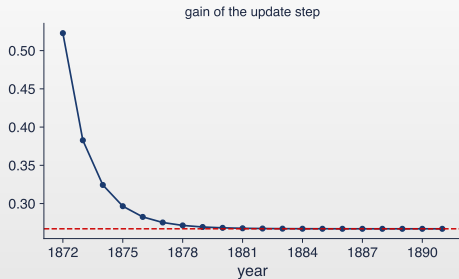


- Annual flow at Aswan, 1871–1970 ( $10^8 \text{ m}^3$ , statsmodels); variances by maximum likelihood with a diffuse start:  
 $\hat{\sigma}_\varepsilon^2 = 15100$ ,  $\hat{\sigma}_\eta^2 = 1468$ ,  $\hat{q} = 0.097$ , the estimates of [Durbin and Koopman \(2012\)](#)

## Interpreting the Nile filter

- Most of the year-to-year movement is noise:  $\hat{q} = 0.097$ , so each flow moves the level by about a quarter of the surprise ( $\bar{K} = 0.267$ )
  - ▶ `statsmodels.UnobservedComponents('llevel')` gives 15078 and 1479: the same model, a slightly different start
- 1899: the flow is 774, the predicted level 1133; the surprise  $v = -359$  lowers the level only to 1037
  - ▶ the filter needs several low years to accept the new level (834 in 1905): the first Aswan dam was completed in 1902
- In 1970 the level is 798 with standard error 63, well below the sample mean 919

## The gain converges; SES reproduces the filter



—●— Kalman gain  $K_t$       • annual flow      - - - SES with  $\alpha = 0.267$   
 - - - steady state  $K = 0.267$       — Kalman one-step forecast

□ Left:  $K_t$  of the Nile filter; right: the Kalman one-step forecasts  $a_t$  and SES with  $\alpha = \bar{K}$ , started at the first observation

## Interpreting the gain

- $K_t$  starts at 0.523 (after the diffuse start), 0.297 in year 5, and is within 0.001 of  $\tilde{K} = 0.267$  after 10 years
  - ▶ the forecasts differ by 0.24 in year 5 and by less than 0.02 after 20 years
- SES fitted directly by least squares ([Chapter 0](#)) gives  $\hat{\alpha} = 0.246$ , i.e.  $q = \alpha^2/(1 - \alpha) = 0.080$ 
  - ▶ close to the likelihood answer: two criteria, one model
- The state space view explains SES:  $\alpha$  is large when the level moves a lot relative to the noise

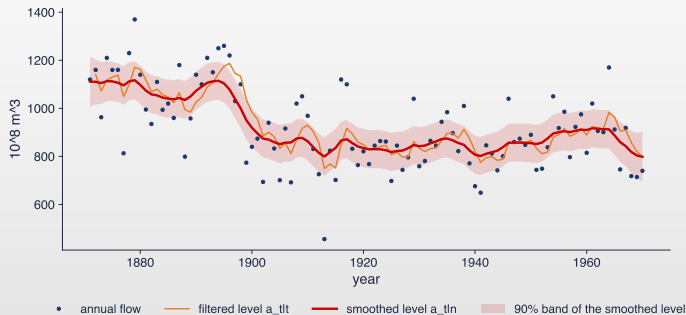
## Recap: The Kalman filter

- Update:  $v_t = y_t - Z_t a_t$ ,  $F_t = Z_t P_t Z_t' + H$ ,  $K_t = P_t Z_t' F_t^{-1}$ ,  $a_{t|t} = a_t + K_t v_t$
- Prediction:  $a_{t+1} = T a_{t|t}$ ,  $P_{t+1} = T P_{t|t} T' + R Q R'$
- The gain weighs the data against the model; it settles at a steady state
- Local level in steady state = SES with  $\alpha = \bar{K}$ ; Nile:  $\bar{K} = 0.267$

## Smoothing: using the whole sample

- ▣ **Filtering** uses  $Y_t$  (the past and the present): the real-time estimate
  - ▶ **smoothing** uses  $Y_n$  (the whole sample):  $\hat{\alpha}_t = E(\alpha_t | Y_n)$ , with variance  $V_t \leq P_{t|t}$
- ▣ **Rauch–Tung–Striebel smoother** Rauch, Tung and Striebel (1965): one backward pass after the filter, from  $t = n - 1$  to 1
  - ▶  $n$ : the sample size;  $J_t$ : the smoother gain, the weight of the revision coming from the future
  - ▶  $J_t = P_{t|t} T' P_{t+1}^{-1}$ ,  $\hat{\alpha}_t = a_{t|t} + J_t(\hat{\alpha}_{t+1} - a_{t+1})$
  - ▶  $V_t = P_{t|t} + J_t(V_{t+1} - P_{t+1})J_t'$ ; at  $t = n$  smoothed and filtered coincide
- ▣ Use: the filtered state for decisions in real time and forecasting; the smoothed state for history (dating breaks, the past output gap)

## The Nile: filtered against smoothed level



□ Filtered level  $a_{t|t}$  and smoothed level  $\hat{a}_t$  with its 90% band; the same maximum-likelihood variances

 TSA\_ch10\_smoothing\_likelihood

## Interpreting the smoothed level

- The smoothed level reacts to the 1898–1899 drop **before** it happens in the data: in 1897 it is already 1038, against 1145 filtered
  - ▶ it knows the future: useful for history, impossible in real time
- In 1900: smoothed 920, filtered 985; the smoothed level adapts faster because it sees the low years after 1900
  - ▶ standard error in 1920: 48 smoothed against 63 filtered
- The smoothed path shows a step near 1899 more clearly than any moving average of [Chapter 0](#)

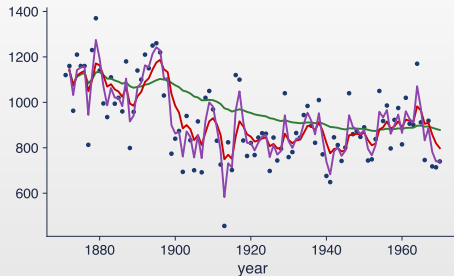
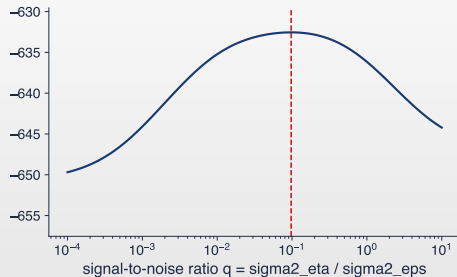
## The likelihood: prediction-error decomposition

- The joint density factorises into one-step conditional densities:  $p(y_1, \dots, y_n) = \prod_t p(y_t | Y_{t-1})$ 
  - ▶ with Gaussian errors,  $y_t | Y_{t-1} \sim N(Z_t a_t, F_t)$ : the filter delivers the mean and the variance
- **Prediction-error decomposition**
  - ▶  $\log L(\theta) = -\frac{n}{2} \log 2\pi - \frac{1}{2} \sum_{t=1}^n \left( \log F_t + \frac{v_t^2}{F_t} \right)$
  - ▶  $\theta$ : the parameters (the variances of the model);  $v_t, F_t$ : the prediction errors and their variances, from the filter
  - ▶ one run of the filter gives  $\log L$  for one value of  $\theta$ ; a numerical optimiser searches over  $\theta$
- Diffuse start: the first  $d$  terms (here  $d = 1$  for the level) are left out, since  $F_t$  is huge there [Durbin and Koopman \(2012\)](#)
- The same idea as the exact ARMA likelihood of [Chapter 2](#), now for any state space model

## Maximum likelihood in practice

- Optimise over  $\log \sigma^2$  (variances stay positive); several starting values, since the surface can be flat
  - ▶ a variance estimated at zero is common and meaningful: that component is deterministic
- **Concentrating**: for the local level, write  $\sigma_\eta^2 = q\sigma_\varepsilon^2$ ; for fixed  $q$ ,  $\hat{\sigma}_\varepsilon^2 = \frac{1}{n-1} \sum_t v_t^2 / F_t$  in closed form
  - ▶ the profile log-likelihood is then a function of  $q$  alone (next chart)
- Standard errors from the numerical Hessian; model choice by AIC and BIC; tests on the standardised prediction errors  $e_t = v_t / \sqrt{F_t}$ 
  - ▶ statsmodels: `UnobservedComponents`, `SARIMAX`, `DynamicFactor` all use this machinery

## Profile likelihood and the choice of $q$



— profile log-likelihood     • annual flow     — filtered level,  $q = 0.097$   
- - - ML:  $q = 0.097$      — filtered level,  $q = 0.001$      — filtered level,  $q = 1$

- Left: profile log-likelihood of the Nile local level model against  $q$  (log scale); right: filtered levels for  $q = 0.001$ , the ML value and  $q = 1$

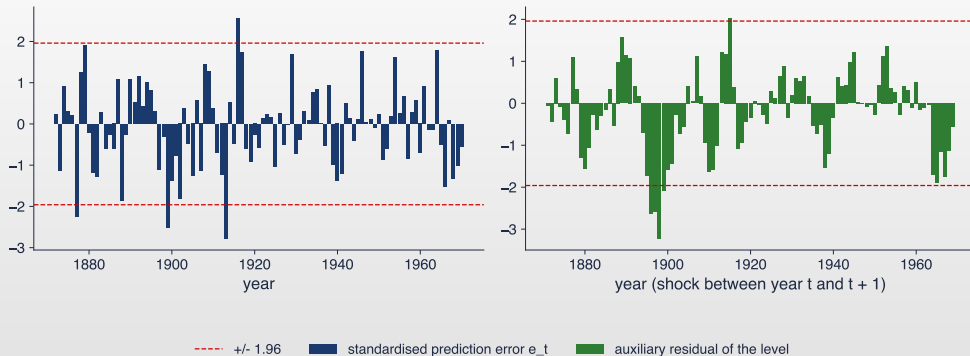
## Interpreting the profile likelihood

- A clear maximum at  $\hat{q} = 0.097$ , with  $\hat{\sigma}_\varepsilon^2 = 15100$  from the closed form
  - ▶ likelihood-ratio statistic against  $q = 0.001$  (an almost constant level): 23.1; against  $q = 1$ : 7.2
- Small  $q$ : the level is almost flat and misses the drop after 1899; large  $q$ : the level chases every flood
  - ▶ the likelihood picks the compromise that predicts best one step ahead
- Choosing  $q$  is the same as choosing the SES weight  $\alpha$  in [Chapter 0](#), now with a statistical criterion

## Diagnostics

- If the model is right, the standardised prediction errors  $e_t = v_t / \sqrt{F_t}$  are i.i.d.  $N(0, 1)$ 
  - ▶ the checks of [Chapters 1–2](#): the ACF and the Ljung–Box test (independence), Jarque–Bera (normality), a plot over time (outliers)
- **Auxiliary residuals** [Durbin and Koopman \(2012\)](#): the smoothed disturbances  $\hat{\varepsilon}_t$  and  $\hat{\eta}_t$ , standardised
  - ▶ a large  $\hat{\varepsilon}_t$ : an **outlier** in one observation
  - ▶ a large  $\hat{\eta}_t$ : a **break** in the level, a permanent shift
- Computed by a backward recursion (the disturbance smoother) after the filter

## Diagnostics of the Nile model



- Left: standardised one-step prediction errors  $e_t$ ; right: standardised auxiliary residual of the level,  $\hat{\eta}_t$  (a shock between year  $t$  and  $t + 1$ ); dashed:  $\pm 1.96$

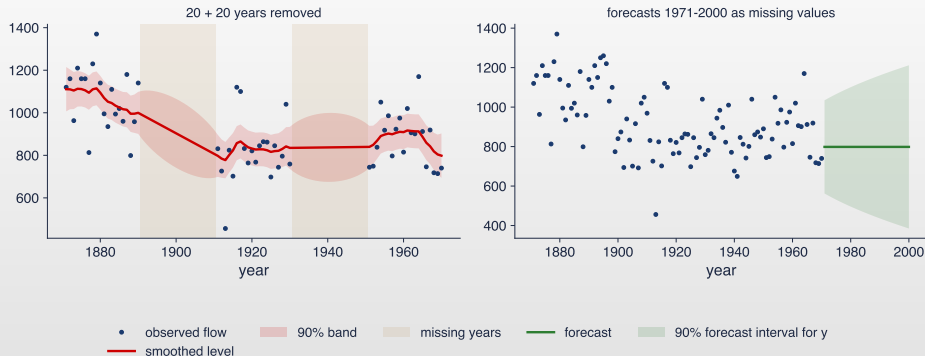
## Interpreting the diagnostics

- Prediction errors: Ljung–Box  $Q(10) = 13.2$  ( $p = 0.21$ ), Jarque–Bera 0.05 ( $p = 0.98$ ); 4 of 99 outside  $\pm 1.96$ 
  - ▶ no evidence against the model in the one-step errors
- The level residual has its minimum in 1898 ( $-3.23$ ): the shift of the level between 1898 and 1899
  - ▶ the observation residual has its minimum in 1913 ( $-3.04$ ): a single very dry year, an outlier
- The same break as in [Chapter 8](#) (spurious long memory of the Nile), now located by the model itself

## Missing values and forecasting

- A missing  $y_t$ : no update;  $a_{t+1} = Ta_t$ ,  $P_{t+1} = TP_tT' + RQR'$ 
  - ▶ the likelihood simply skips that term; nothing is interpolated by hand
- Forecasting  $h$  steps ahead = treating  $y_{n+1}, \dots, y_{n+h}$  as missing
  - ▶ local level:  $\hat{y}_{n+h} = a_{n+1}$  for every  $h$ , with variance  $P_{n+1} + (h-1)\sigma_\eta^2 + \sigma_\varepsilon^2$
- Uses: irregular data (holidays), series that start at different dates, mixed frequencies, the ragged edge of a nowcast
  - ▶ the smoother fills the gaps with  $\hat{\alpha}_t$  and the corresponding standard error

## Missing years and forecasts of the Nile



- Left: 1891–1910 and 1931–1950 removed (the experiment of [Durbin and Koopman \(2012\)](#)), smoothed level with 90% band; right: forecasts for 1971–2000 with 90% intervals for  $y$

## Interpreting the missing data

- Inside a gap the smoothed level is a straight line between the two edges; its standard error grows to 99 in 1900, against 48 with all data
  - ▶ the band widens towards the middle of the gap and narrows near observed years
- Forecasts are flat at 798: a random-walk level has no predictable direction
  - ▶ standard error of  $y$ : 144 one year ahead, 251 after 30 years
- The same code handles missing data, forecasts and estimation: one of the main practical strengths of state space models

## Recap: Smoothing, likelihood and missing data

- ▣ Filtered: real time; smoothed (RTS): the whole sample, smaller variance
- ▣  $\log L = -\frac{1}{2} \sum (\log 2\pi + \log F_t + v_t^2/F_t)$ , maximised numerically
- ▣ Diagnostics on  $e_t$ ; auxiliary residuals find outliers and breaks (Nile 1898–1899)
- ▣ Missing data and forecasts: skip the update step

## Unobserved components models

- ▣ **Structural time series** or **unobserved components (UC)** model [Harvey \(1989\)](#):  $y_t = \mu_t + \psi_t + \gamma_t + \varepsilon_t$ 
  - ▶ trend  $\mu_t$  (local level or local linear trend), cycle  $\psi_t$ , seasonal  $\gamma_t$ , irregular  $\varepsilon_t$
  - ▶ the decomposition of [Chapter 0](#), but each component is a stochastic process with its own variance
- ▣ The **cycle**: a stationary AR(2)  $\psi_t = \phi_1\psi_{t-1} + \phi_2\psi_{t-2} + \kappa_t$ , or a damped stochastic cycle with period  $2\pi/\lambda_c$ 
  - ▶  $\phi_1, \phi_2$ : the AR coefficients of the cycle;  $\kappa_t$ : the cycle shock;  $\lambda_c$ : the frequency of the cycle, in radians
  - ▶ for log GDP,  $\psi_t$  is the **output gap**: the percentage deviation of output from its trend (potential)
- ▣ In Python: `UnobservedComponents(y, level='smooth trend', autoregressive=2)`; the whole state is estimated by the Kalman filter and smoother

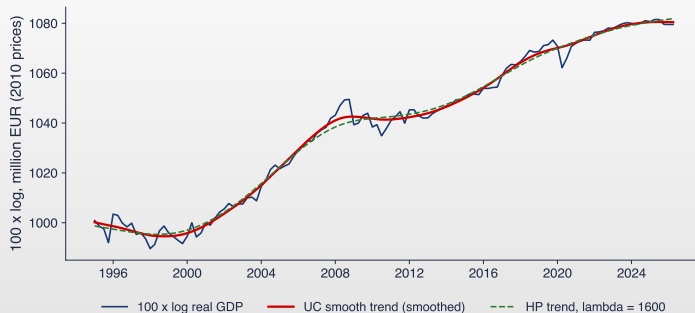
## The HP filter is a state space smoother

- ▣ Hodrick and Prescott (1997): the trend  $\tau_t$  minimises  $\sum_t (y_t - \tau_t)^2 + \lambda \sum_t (\Delta^2 \tau_t)^2$ ;  $\lambda = 1600$  for quarterly data
  - ▶  $\Delta^2 \tau_t = \tau_t - 2\tau_{t-1} + \tau_{t-2}$ : the change in the slope of the trend;  $\lambda \geq 0$ : the smoothing parameter
  - ▶ large  $\lambda$ : a straight-line trend;  $\lambda = 0$ : the trend is the series itself
- ▣ This is exactly the **smoothed** trend of a UC model: smooth trend ( $\Delta^2 \mu_{t+1} = \zeta_t$ ) plus white noise, with  $\lambda = \sigma_\varepsilon^2 / \sigma_\zeta^2$  Harvey and Jaeger (1993)
  - ▶ the HP filter imposes the signal-to-noise ratio and a white-noise gap; the UC model estimates both
- ▣ Being a smoother, HP uses future data: its last values are revised as new quarters arrive

## Hamilton (2018): why you should never use the HP filter

- Hamilton (2018) lists three problems
  - ▶ the HP gap of a random walk has cycles that are not in the data: spurious dynamics
  - ▶ the end of the sample is treated differently from the middle: large revisions, misleading real-time gaps
  - ▶  $\lambda = 1600$  is a convention, far from the value a likelihood would choose
- His alternative, the **regression filter**: OLS of  $y_{t+h}$  on  $1, y_t, y_{t-1}, y_{t-2}, y_{t-3}$  with  $h = 8$  quarters
  - ▶ the cycle is the residual: what could not be predicted two years earlier; one-sided by construction
- No method is the truth: we compare the UC model, HP and the Hamilton filter on Romanian GDP

## Romanian GDP: trend

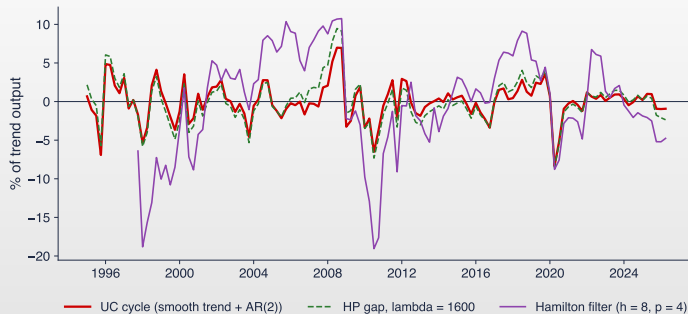


- 100 × log of real GDP, quarterly, seasonally and calendar adjusted, chain-linked volumes (Eurostat namq\_10\_gdp), 1995Q1–2026Q2 ( $T = 126$ ); UC model: smooth trend plus AR(2) cycle; HP with  $\lambda = 1600$

## Interpreting the trend

- ▣ Average growth 2.5% per year, with a slow start (the 1997–1999 recession), a boom until 2008, a stagnation and a recovery after 2013
  - ▶ the two trends are almost identical: the difference between methods lies in the gap, not in the trend
- ▣ UC estimates:  $\hat{\sigma}_{\zeta}^2 = 0.078$  (trend slope), cycle AR(2) with  $\hat{\phi}_1 = 0.61$ ,  $\hat{\phi}_2 = -0.13$ ,  $\hat{\sigma}_{\kappa}^2 = 5.27$ ; the irregular variance is estimated at zero
  - ▶ the ratio cycle variance / trend variance is 68, far below the 1600 that HP imposes: a more flexible trend
- ▣ A flexible trend absorbs part of the boom: the UC gap will be smaller than the HP gap

## The Romanian output gap: three methods

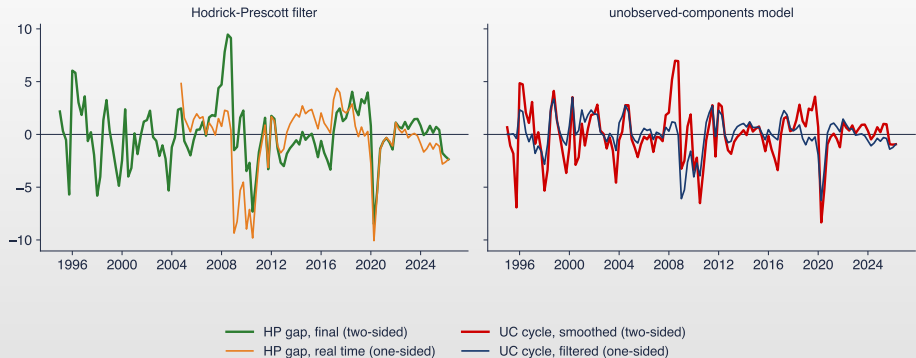


- Gap in % of trend: UC cycle (smoothed), HP gap ( $\lambda = 1600$ ) and the Hamilton filter ( $h = 8$ , four lags), the first 11 quarters lost

## Interpreting the output gaps

- All three see the overheating before 2008 (peaks: UC 7.0% in 2008Q3, HP 9.5%, Hamilton 10.7%) and the 2020 pandemic (UC  $-8.3\%$ , HP  $-8.5\%$ )
  - ▶ correlations: UC–HP 0.94, UC–Hamilton 0.57, HP–Hamilton 0.69
- Standard deviations: UC 2.4, HP 2.9, Hamilton 6.5: the Hamilton gap also contains every surprise of two years
  - ▶ the four regression coefficients sum to 0.97: GDP is close to a random walk with drift
- Latest gap (2026Q2): UC  $-0.9\%$ , HP  $-2.4\%$ , Hamilton  $-4.8\%$ : the sign agrees, the size does not

## Real time against hindsight

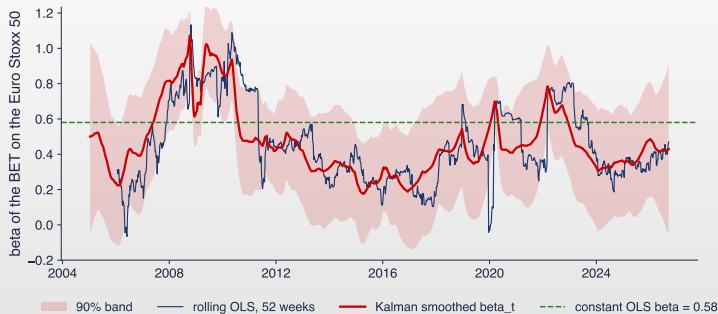


- Left: the HP gap computed with all data (two-sided) and the last value of the HP gap computed with the data available at each date (one-sided, from 2004); right: the UC cycle, filtered and smoothed

## Interpreting the revisions

- In 2008Q3 the real-time HP gap was 2.2%; with hindsight it is 9.5%: the overheating was invisible in real time
  - ▶ the UC model shows the same problem: filtered 1.1%, smoothed 7.0%
- Root mean square revision since 2004: HP 2.8 points, UC 1.8 points; correlation real time–final: HP 0.57, UC 0.65
  - ▶ the end-point problem of [Hamilton \(2018\)](#); the UC model at least reports its own uncertainty
- For policy in real time, use the filtered (one-sided) estimate and its standard error, never the last point of a two-sided filter

## A time-varying beta: the BET and the euro area



- Weekly log returns of the BET on the Euro Stoxx 50 (EODHD), 2005–2026 ( $T = 1\,126$  weeks); random-walk beta by maximum likelihood, Kalman smoothed, with a 90% band; 52-week rolling OLS; constant OLS

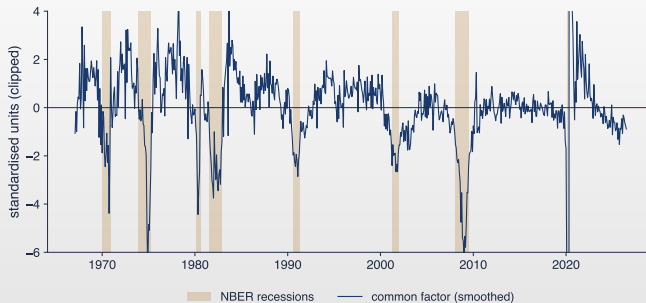
## Interpreting the time-varying beta

- ▣ The constant OLS beta is 0.58; the smoothed beta peaks at 1.07 (October 2008) and falls to 0.17 (April 2015)
  - ▶ in the 2008 crisis the Romanian market moved almost one for one with the euro area; later it decoupled
- ▣ Likelihood ratio against a constant beta: 46.1 (one restriction, a variance on the boundary): strong evidence of variation
  - ▶ weekly standard deviation of the beta shocks 0.059; latest beta 0.43 with standard error 0.29
- ▣ The rolling window jumps when a single extreme week enters or leaves it; the Kalman beta moves only as much as the likelihood allows

## Dynamic factor models and nowcasting

- **Dynamic factor model:**  $y_{it} = \lambda_i f_t + u_{it}$ ,  $f_t = \phi_1 f_{t-1} + \phi_2 f_{t-2} + \eta_t$ 
  - ▶ many series  $y_{1t}, \dots, y_{Nt}$  driven by one unobserved common factor  $f_t$  (the state) with loadings  $\lambda_i$
  - ▶  $u_{it}$ : the part of series  $i$  not explained by the factor;  $\phi_1, \phi_2$ : the AR coefficients of the factor
  - ▶ **Stock and Watson (1989)**: a coincident index of US activity from four monthly indicators
- **Nowcasting** **Giannone, Reichlin and Small (2008)**: estimate the current quarter before GDP is published
  - ▶ monthly data arrive at different dates: the latest months of some series are missing (the **ragged edge**)
  - ▶ the Kalman filter treats them as missing values and updates the factor with whatever has arrived
- Central banks use such models to nowcast GDP; statsmodels has `DynamicFactor` and `DynamicFactorMQ`

## A common factor of US activity



- Monthly growth of industrial production, payroll employment, real income less transfers and real manufacturing and trade sales (FRED), standardised; one AR(2) factor estimated on 1967–2019 and run to July 2026, the last two months of income and sales set to missing

## Interpreting the common factor

- ▣ Loadings: employment 0.59, industrial production 0.42, sales 0.29, income 0.23; factor AR(2) with  $\hat{\phi}_1 = 0.42$ ,  $\hat{\phi}_2 = 0.41$ 
  - ▶ correlation with the simple average of the four series: 0.86
- ▣ Mean of the factor:  $-3.5$  in NBER recession months,  $0.4$  in expansions; negative in 99% of recession months
  - ▶ April 2020:  $-74$  standard units, off the scale of the chart
- ▣ Latest value (July 2026):  $-0.9$ , computed although two of the four series are not yet available

## Recap: Trend, cycle and common factors

- ▣ UC models: trend + cycle + seasonal + irregular, each with its own variance
- ▣ HP = smoother of a UC model with imposed  $\lambda$ ; Hamilton (2018): spurious cycles and end-point revisions
- ▣ Romanian output gap: the methods agree on the sign, not on the size; real-time gaps are revised heavily
- ▣ TVP regression and dynamic factors are state space models too

## Hamilton (1989) and the dating of recessions



The NBER, Cambridge (Massachusetts), whose committee dates US recessions

Photo: [Astrophobe \(2022\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

- The National Bureau of Economic Research (NBER) dates US recessions by a committee, months after the fact
  - ▶ a recession is a regime: GDP behaves differently while it lasts
- [Hamilton \(1989\)](#): let the mean of GDP growth depend on an unobserved regime  $S_t \in \{1, 2\}$  that follows a Markov chain
  - ▶ the data then tell us the probability of a recession in each quarter, without a committee
- The state is now **discrete**; the Kalman filter is replaced by the Hamilton filter

## The Markov-switching model

- **Switching mean:**  $y_t = \mu_{S_t} + \varepsilon_t$ ,  $\varepsilon_t \sim N(0, \sigma_{S_t}^2)$ ,  $S_t \in \{1, \dots, k\}$ 
  - ▶ **Markov chain:**  $p_{ij} = \Pr(S_t = j \mid S_{t-1} = i)$ , constant over time; the rows of the transition matrix sum to 1
- Hamilton's MS-AR(4):  $y_t - \mu_{S_t} = \sum_{j=1}^4 \phi_j (y_{t-j} - \mu_{S_{t-j}}) + \varepsilon_t$ 
  - ▶ the AR dynamics stay the same; only the mean switches
- Variants: switching variance (calm and turbulent markets), switching AR coefficients,  $k = 3$  regimes
  - ▶ in Python: `MarkovRegression`, `MarkovAutoregression` (`statsmodels`)
- A nonlinear model: the forecast depends on the probability of each regime

## Transition matrix, durations and ergodic probabilities

- Two regimes:  $P = \begin{pmatrix} p_{11} & 1 - p_{11} \\ 1 - p_{22} & p_{22} \end{pmatrix}$ 
  - ▶  $h$ -step transitions:  $P^h$
- **Expected duration** of regime  $i$ : the time spent there is geometric,  $\Pr(D = d) = p_{ii}^{d-1}(1 - p_{ii})$ , so  $E(D) = 1/(1 - p_{ii})$ 
  - ▶  $p_{11} = 0.75$ : recessions last  $1/0.25 = 4$  quarters on average;  $p_{22} = 0.95$ : expansions last 20 quarters
- **Ergodic** (long-run) probabilities:  $\pi_1 = \frac{1 - p_{22}}{2 - p_{11} - p_{22}}$ , the share of time spent in regime 1
  - ▶ example:  $\pi_1 = 0.05/0.30 = 1/6$ : one quarter in six is a recession quarter

## The Hamilton filter

- Track  $\xi_{t|t} = \Pr(S_t = j \mid Y_t)$ , the **filtered probabilities**, for each regime  $j$ 
  - ▶ **prediction**:  $\Pr(S_t = j \mid Y_{t-1}) = \sum_i p_{ij} \Pr(S_{t-1} = i \mid Y_{t-1})$
  - ▶ **densities**:  $f_j(y_t) = \phi((y_t - \mu_j)/\sigma_j)/\sigma_j$ , the likelihood of  $y_t$  in each regime;  $\phi$ : the standard Normal density;  $\mu_j, \sigma_j$ : the mean and standard deviation of regime  $j$
  - ▶ **update** (Bayes):  $\Pr(S_t = j \mid Y_t) = \frac{\Pr(S_t = j \mid Y_{t-1}) f_j(y_t)}{\sum_i \Pr(S_t = i \mid Y_{t-1}) f_i(y_t)}$
- The denominator is  $p(y_t \mid Y_{t-1})$ : the log-likelihood is  $\sum_t \log p(y_t \mid Y_{t-1})$ , as for the Kalman filter
  - ▶ the same predict–update cycle, with probabilities instead of means and variances
- Start: the ergodic probabilities  $\pi_j$

## Smoothed probabilities and estimation

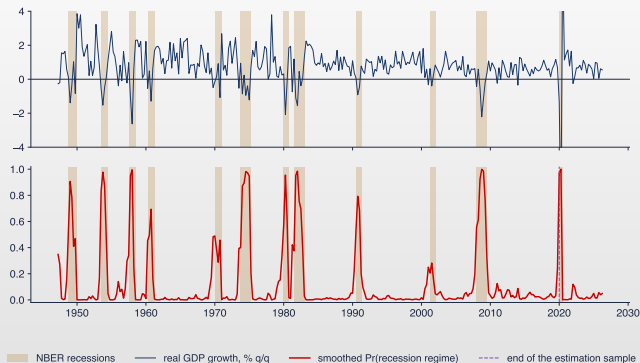
- ▣ **Smoothed probabilities**  $\Pr(S_t = j \mid Y_n)$ : a backward pass [Kim \(1994\)](#), the analogue of the RTS smoother
  - ▶ used to date regimes in history; the filtered probabilities are what a forecaster knew at time  $t$
- ▣ Estimation: maximum likelihood by numerical optimisation, or the EM algorithm [Hamilton \(1990\)](#)
  - ▶ EM: expectation (smoothed probabilities given  $\theta$ ) and maximisation (weighted means, variances, transition counts), repeated
- ▣ The likelihood has **several local maxima**: use many random starting values and compare the solutions
  - ▶ label switching: regime 1 and regime 2 can swap; name them by their means or variances

## Hamilton's specification on today's data

| MS-AR(4), US real GDP growth,<br>1951Q2–1984Q4          | Maximum A    | Maximum B    |
|---------------------------------------------------------|--------------|--------------|
| mean, regime 1 / regime 2 (% per quarter)               | −0.14 / 1.32 | −1.14 / 0.96 |
| $p_{11}$ , $p_{22}$                                     | 0.762, 0.892 | 0.128, 0.952 |
| expected duration, regime 1 / 2 (quarters)              | 4.2 / 9.3    | 1.1 / 20.8   |
| log-likelihood                                          | −189.47      | −189.16      |
| NBER recession quarters found (smoothed prob.<br>> 0.5) | 94%          | 10%          |

- Maximum A is Hamilton's answer: a recession regime of about four quarters that matches the NBER dates
- Maximum B has a slightly higher likelihood but a different meaning: isolated sharp quarters
- A likelihood difference of −189.16 against −189.47 cannot decide; economic meaning and robustness checks must

## US recessions from GDP growth

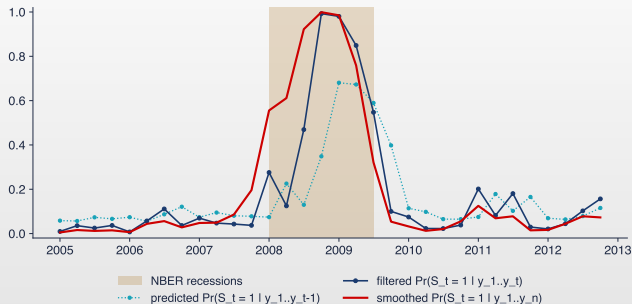


- Two regimes with switching mean, constant variance, estimated on 48 NBER recession quarters out of 1947Q2–2019Q4 (FRED GDPC1, USREC); probabilities after 2019 computed with the same parameters

## Interpreting the recession probabilities

- ▣ Regime means:  $-0.42\%$  (recession) and  $0.97\%$  (expansion) per quarter,  $\hat{\sigma} = 0.79$ ;  $\hat{p}_{11} = 0.685$ ,  $\hat{p}_{22} = 0.949$ 
  - ▶ expected durations: 3.2 quarters of recession, 19.7 quarters of expansion; ergodic share of recession 14%
- ▣ Agreement with the NBER: 94% of quarters; 62% of NBER recession quarters detected, 0% false alarms
  - ▶ the model misses the mild recessions (2001) and catches the deep ones
- ▣ 2020Q2 ( $-8.2\%$ ): probability 1.00, then back to expansion; after 2021 the probability never exceeds 0.12

## Filtered against smoothed probabilities: 2008



□ Predicted, filtered and smoothed probability of the recession regime, 2005–2012, same model as on the previous chart

 TSA\_ch10\_markov\_gdp

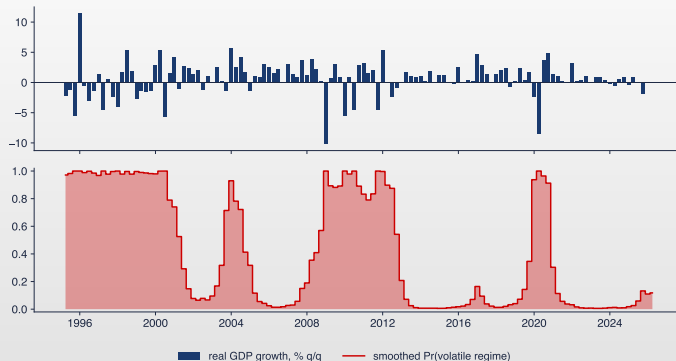
## Interpreting the real-time probabilities

- The NBER recession started in 2008Q1; the smoothed probability crosses 0.5 in 2008Q1, the filtered one only in 2008Q4
  - ▶ 2008Q1: filtered 0.28, smoothed 0.56; 2008Q3 (growth  $-0.53\%$ ): filtered 0.47, smoothed 0.92
- Concordance with the NBER using only filtered probabilities: 92% of quarters, 54% of recession quarters
  - ▶ real-time dating is harder: [Chauvet and Piger \(2008\)](#) compare the methods on real-time data
- Never judge a model's real-time skill from its smoothed probabilities: they use the future

## Checks: which regimes did the model find?

- ▣ **Switching variance** on 1947–2019: the regimes become high and low volatility, not recession and expansion
  - ▶ standard deviations 1.20 and 0.44; durations 45 and 39 quarters; the switch to the calm regime in 1984Q3
  - ▶ this is the “Great Moderation” of the mid-1980s; agreement with the NBER falls to 60% (47% false alarms)
- ▣ **Including 2020** in the estimation: one regime is the single quarter 2020Q2 (mean  $-8.2\%$ ), the other lasts 315 quarters
  - ▶ one extreme observation can capture a whole regime
- ▣ Checklist: plot the probabilities against known events; compare several starting values and samples; prefer the specification with a clear economic meaning

## Romanian GDP growth regimes

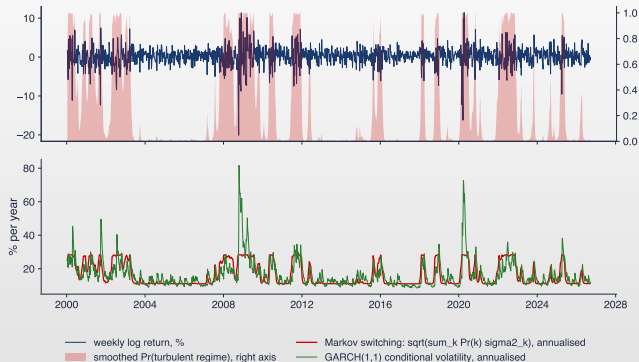


- Quarter-on-quarter growth of Romanian real GDP (Eurostat), two regimes with switching mean and variance; smoothed probability of the volatile regime

## Interpreting the Romanian regimes

- Stable regime: mean 1.03% per quarter, standard deviation 1.3; volatile regime: mean 0.07%, standard deviation 3.9
  - ▶ expected durations: 15.2 quarters stable, 11.2 quarters volatile; 40% of the quarters are volatile
- Volatile spells: 1995Q2–2001Q2 (transition and the 1997–1999 recession), 2003Q4–2004Q3, 2008Q4–2012Q4 (the crisis and the austerity of 2010), 2020Q1–2020Q4 (the pandemic)
  - ▶ latest probability of the volatile regime: 0.12
- With 125 quarters a mean-only model is unstable; regimes of volatility and growth come together in an emerging economy

## Calm and turbulent stock markets



- Weekly log returns of the S&P 500, 2000–2026 (EODHD); two regimes with switching mean and variance; bottom: the volatility implied by the regime probabilities against a GARCH(1,1) (Chapter 5), both annualised

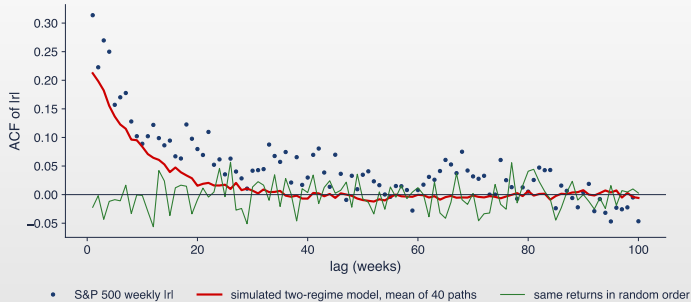
## Interpreting the volatility regimes

- Calm regime: volatility 11% per year, mean 0.32% per week; turbulent: 28% per year, mean  $-0.39\%$  per week
  - ▶ durations: 37 weeks calm, 14 weeks turbulent; turbulent in 26% of the weeks (ergodic 28%)
- Most turbulent years: 2022 (90% of the weeks), 2008 (79%), 2009 (69%)
  - ▶ correlation with the GARCH(1,1) volatility: 0.68; GARCH:  $\hat{\alpha} = 0.26$ ,  $\hat{\beta} = 0.70$
- Two views of the same clustering: GARCH moves volatility continuously, the regime model jumps between two levels

## Regimes, GARCH persistence and long memory

- Lamoureux and Lastrapes (1990): shifts in the level of variance make a GARCH look almost integrated ( $\alpha + \beta \approx 1$ )
  - ▶ the near-IGARCH estimates of Chapter 5 may reflect regimes, not one very persistent process
  - ▶ Hamilton and Susmel (1994): ARCH within regimes (SWARCH) lowers the persistence
- Diebold and Inoue (2001): rare regime switches produce a slowly decaying ACF and  $\hat{d} > 0$  (Chapter 8, spurious long memory)
  - ▶ three models, one fact: volatility shocks persist; the next chart simulates the fitted regime model to see how far regimes alone go

## Regimes imitate memory



- ACF of weekly  $|r_t|$  of the S&P 500; mean ACF of 40 series simulated from the fitted two-regime model (constant variance within each regime, no GARCH); the same returns in random order

## Interpreting the simulated regimes

- Data: ACF of  $|r_t|$  0.31 at lag 1, 0.06 at 26 weeks; regime model: 0.21 and 0.01; random order: none
  - ▶ the regimes explain most of the short-run clustering, less of the long tail
- Local Whittle  $\hat{d}$  of  $|r_t|$ : 0.32 in the data, 0.29 (SD 0.06) in the simulated regime series,  $-0.05$  in random order
  - ▶ a GARCH(1,1) fitted to the simulated series gives  $\alpha + \beta = 0.91$  on average (15 series), although there is no GARCH in them
- Persistence and long memory in volatility are consistent with regimes: compare models by likelihood and forecasts, not by one statistic

## Recap: Markov switching

- ▣  $y_t = \mu_{S_t} + \varepsilon_t$ ,  $S_t$  a Markov chain; durations  $1/(1 - p_{ii})$ ; ergodic  $\pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$
- ▣ Hamilton filter: predict with P, update with Bayes; Kim smoother for history
- ▣ US GDP: the model recovers the deep NBER recessions; filtered probabilities react later
- ▣ Several maxima, outliers and variance breaks can change what a “regime” means
- ▣ Regimes reproduce volatility clustering, GARCH persistence and apparent long memory

## Possible contribution of AI

- ▣ **Code:** a first draft of a Kalman filter, of the matrices of a new model, of a Markov-switching fit with many starting values
- ▣ **Explanation:** a second derivation of the gain, of the steady state, of the Hamilton filter
- ▣ **Exploration:** output gaps for many EU countries; regimes in many markets; a nowcast with many indicators
- ▣ Example prompt
  - ▶ Write Python code that downloads Romanian quarterly real GDP from Eurostat (`namq_10_gdp, Q.CLV10_MEUR.SCA.B1GQ.RO`), fits `UnobservedComponents` with a smooth trend and an `AR(2)` cycle, compares the smoothed and the filtered cycle with the HP gap and with the Hamilton (2018) regression filter, and reports the revisions of the last eight quarters.

## Checks you must run

- ▣ Simulate from known parameters and check that the code recovers them (the Nile values of [Durbin and Koopman \(2012\)](#) are a good test)
- ▣ Filtered or smoothed? Real-time claims need filtered estimates; an AI answer that uses smoothed for a “forecast” is wrong
- ▣ Durations:  $1/(1 - p_{ii})$ , not  $p_{ii}/(1 - p_{ii})$ ; check the convention of the transition matrix (rows or columns) in the library
- ▣ Markov switching: several starting values, label switching, outliers that capture a regime
- ▣ The HP filter at the end of the sample: do not read the last gap as a real-time estimate
- ▣ Every cited reference: it must exist; check the DOI

## Key takeaways

- ▣ State space form: a hidden state  $\alpha_t$  follows a VAR(1) and is measured with noise; ARMA, ETS, UC, TVP and factor models fit in it
- ▣ The Kalman filter predicts and updates; the gain weighs data against model; the local level filter in steady state is SES
- ▣ Smoothing for history, filtering for real time; the likelihood comes from the prediction errors; missing data are easy
- ▣ Output gaps depend on the method and are revised heavily at the end of the sample
- ▣ Markov switching: regime probabilities, durations  $1/(1 - p_{ii})$ ; check what the regimes mean

## Key formulas

| Quantity                 | Formula                                                                                                                           |
|--------------------------|-----------------------------------------------------------------------------------------------------------------------------------|
| State space form         | $y_t = Z_t \alpha_t + \varepsilon_t, \quad \alpha_{t+1} = T \alpha_t + R \eta_t$                                                  |
| Update                   | $v_t = y_t - Z_t a_t, \quad F_t = Z_t P_t Z_t' + H, \quad K_t = P_t Z_t' F_t^{-1}, \quad a_{t t} = a_t + K_t v_t$                 |
| Prediction               | $a_{t+1} = T a_{t t}, \quad P_{t+1} = T P_{t t} T' + R Q R'$                                                                      |
| Local level steady state | $\bar{P} = \sigma_\varepsilon^2 (q + \sqrt{q^2 + 4q})/2, \quad \alpha_{SES} = \bar{K} = \bar{P}/(\bar{P} + \sigma_\varepsilon^2)$ |
| Log-likelihood           | $-\frac{1}{2} \sum_t (\log 2\pi + \log F_t + v_t^2/F_t)$                                                                          |
| HP filter                | $\min \sum (y_t - \tau_t)^2 + \lambda \sum (\Delta^2 \tau_t)^2, \quad \lambda = \sigma_\varepsilon^2 / \sigma_\zeta^2$            |
| Markov chain             | $E(D_i) = 1/(1 - p_{ii}), \quad \pi_1 = (1 - p_{22})/(2 - p_{11} - p_{22})$                                                       |
| Hamilton filter          | $\Pr(S_t = j \mid Y_t) \propto f_j(y_t) \sum_i p_{ij} \Pr(S_{t-1} = i \mid Y_{t-1})$                                              |

## Self-assessment

- **Question:** local level with  $P_t = 3$ ,  $\sigma_\varepsilon^2 = 1$ . What is the Kalman gain?  
▶ **Answer:**  $K_t = 3/(3 + 1) = 0.75$ : the new observation gets weight 0.75
- **Question:** an SES model has  $\alpha = 0.5$ . Which local level model is behind it?  
▶ **Answer:**  $q = \alpha^2/(1 - \alpha) = 0.25/0.5 = 0.5$
- **Question:**  $p_{11} = 0.9$  in a two-regime model. How long does regime 1 last on average?  
▶ **Answer:**  $1/(1 - 0.9) = 10$  periods
- Next: [Chapters 11–14](#) for self-study; [Chapter 15](#), review

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